

Bonding and Electronegativity

V · T · E

Bonding of simple substances in the periodic table

[hide]

	1	2											3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18				
Group →																																
↓ Period																																
1	H																												He			
2	Li	Be																						B	C	N	O	F	Ne			
3	Na	Mg																						Al	Si	P	S	Cl	Ar			
4	K	Ca											Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr				
5	Rb	Sr											Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe				
6	Cs	Ba	La	Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
7	Fr	Ra	Ac	Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	Rf	Db	Sg	Bh	Hs	Mt	Ds	Rg	Cn	Nh	Fl	Mc	Lv	Ts	Og

Metallic

Network covalent

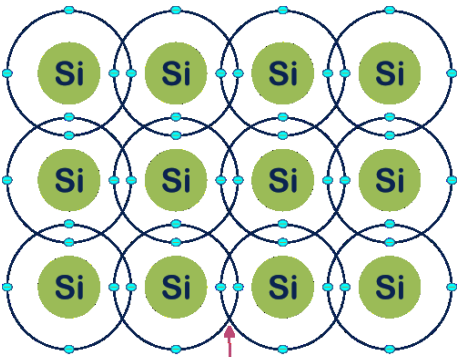
Molecular covalent

Single atoms

Unknown

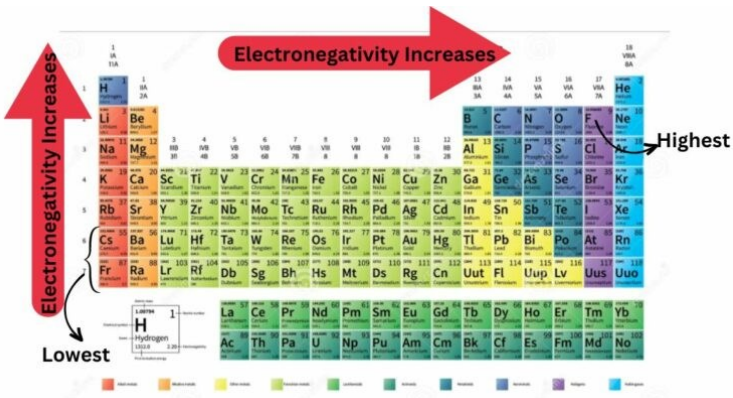
Background color shows bonding of simple substances in the periodic table. If there are several, the most stable allotrope is considered.

[https://en.wikipedia.org/wiki/Template:Periodic_table_\(simple_substance_bonding\)](https://en.wikipedia.org/wiki/Template:Periodic_table_(simple_substance_bonding))



Covalent bond

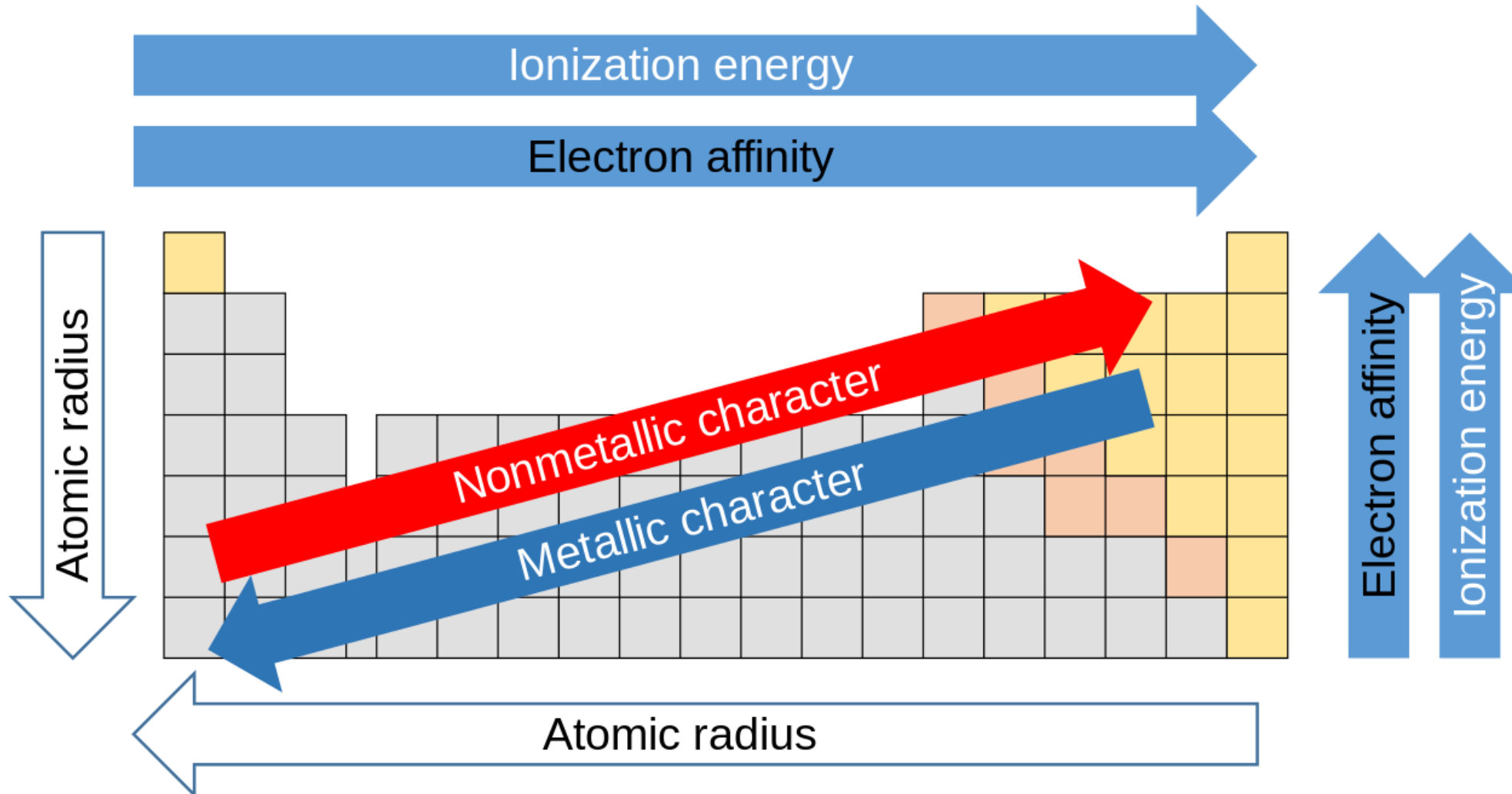
<https://vedify.com/wp-content/uploads/2023/09/Silicon-Si-atoms-covalent-bond-with-the-neighbour-Si-atom.png>



<https://scienceinfo.com/wp-content/uploads/2023/08/Electronegativity-Increases-768x402.jpg>

Electronegativity: tendency to attract electrons

Other Trends in the Periodic Table



[https://commons.wikimedia.org/wiki/File:Periodic trends %282%29.svg](https://commons.wikimedia.org/wiki/File:Periodic_trends_%282%29.svg)

- **Analogy to water dispenser**

- A water dispenser allows water to pass through it when a mechanism is used, e.g., a button is pressed.
- The electrons of a semiconductor move to the conduction band and the relevant (semiconductor) material conducts when enough energy is provided.

- **Overall functionality**

- In this way, the movement of charge can be controlled, leading to individual states both of the semiconductor (i.e., its conducting and its non-conducting state) and the other *electrical* components that constitute the electronic device (e.g., a capacitor being charged or discharged).
- This, in turn, means that electronics can be connected to binary states and the Boolean algebra. Nevertheless, electronics with more than two states can be utilized for unconventional computing, e.g., for ternary computing.

- **Analogy to water**

- Conductor (electron cloud): Canal
- Semiconductor (many covalent bonds): Filled up canal/lake/pool at the ground level
- Insulator (a few covalent bonds): Filled up canal with dams of rocks
- Negative electrode of a battery: Filled up lake/pool at a high level
- Positive electrode of a battery: Empty lake/pool below ground level
- Ground: Sea
- Power plug at the wall: Huge mountain lake/sea



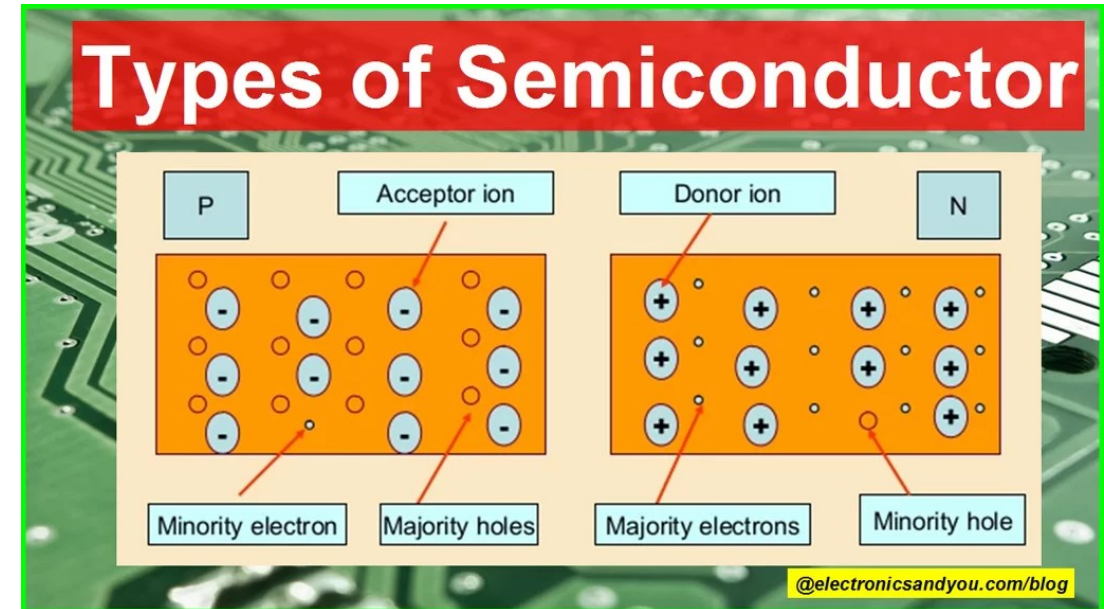
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<https://assets.thehansindia.com/h-upload/2021/09/04/1106050-river-waters.webp>

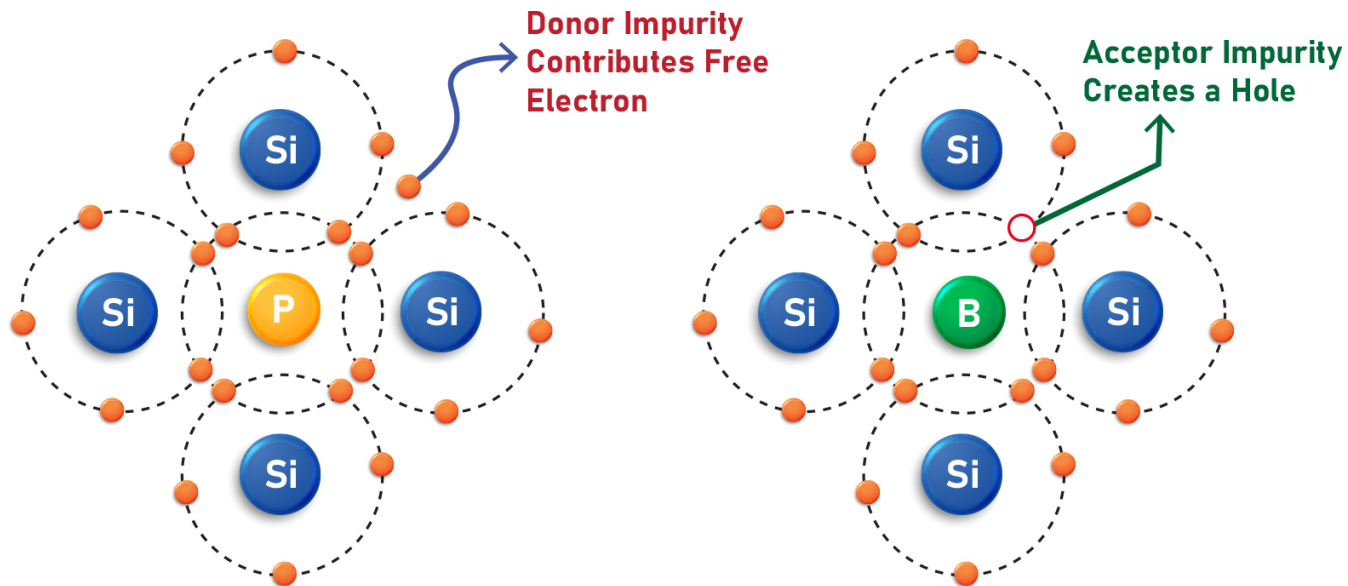
Doped/Extrinsic Semiconductors

- Silicon is a semiconductor on its own. Thus, it is an intrinsic semiconductor.
- However, in order to get more distinguishable states, it sometimes makes sense to enhance the number of electrons that can reach the conduction band, by increasing the number of electrons or electron holes.
- This can be done by introducing into the crystal lattice structure of silicon, atoms of elements with more electrons in their valence band (electron donors), such as P atoms, or with fewer electrons (and thus more (electron) holes) in their valence band (electron acceptors), such as B atoms, respectively.
- These “impurities” will lead to an abundance of electrons or (electron) holes in the (regional) valence bands, respectively.
- As electrons are negatively charged, the former form of “doping” leads to a more negatively charged (n-type) semiconductor and the latter “doping” to a more positively charged (p-type) semiconductor, respectively.



<https://www.electronicsandyou.com/blog/semiconductor.html>

Extrinsic Semiconductor



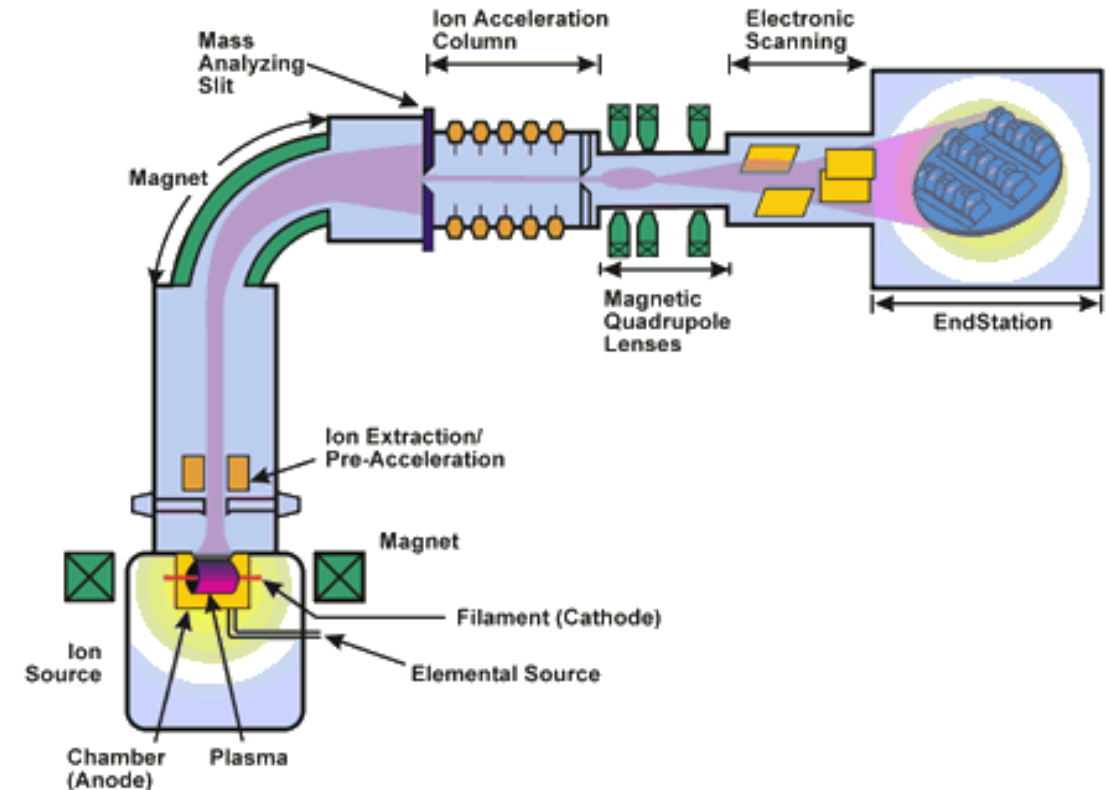
N-Type Semiconductor

P-Type Semiconductor

<https://images.theengineeringprojects.com/image/webp/2023/09/intrinsic-semiconductor-02.jpg.webp?ssl=1>

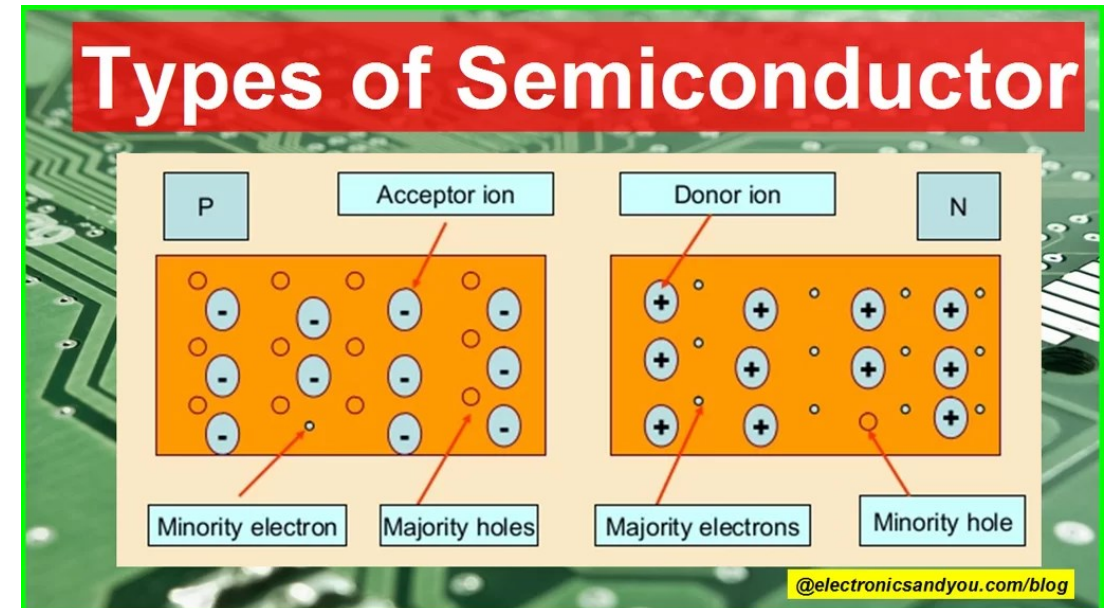
Doped/Extrinsic Semiconductors

- Main way of doping silicon: Ion implantation
- Electron gun / particle beam device, releases small amounts of vaporized phosphorus (P), arsenic (As) or boron (B), by accelerating the resulting ions to a high velocity and ramming them into specific positions of the silicon wafer.
- Thermal Diffusion: Heating the silicon wafers in a closed tube, pumping in some vaporized phosphorus (P), arsenic (As) or boron (B). The dopants diffuse into the solid silicon.
- Until the late 1960s, diffusion was the primary method. In the 1970s and 1980s, ion implantation became dominant for all but creating the doping of bulk wafers.
- Ion implantation is much more precise but it also causes damage to the silicon crystal structure. The “damage” needs thermal annealing in order to make the impurities (dopants) fit into the silicon crystal structure (through bonding).
- One can control in which region the dopants are deposited by masking in the photolithography process.

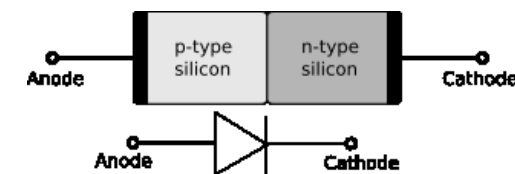


<https://qph.cf2.quoracdn.net/main-qimg-3828444366cd59699e509f3170f316ee>

- Since both n-type and p-type semiconductors have an abundance of free electrons, it will be easy for electrical current to flow through them.
- It is rather interesting, however, to examine what happens when an n-type semiconductor and a p-type semiconductor are placed next to each other (contacting each other), for example as an n-doped silicon region next to a p-doped silicon region, in order to form what we call a p-n junction.
- In this case, the p-type semiconductor forms the anode and the n-type semiconductor the cathode of the component, which is essentially a diode.



<https://www.electronicsandyou.com/blog/semiconductor.html>

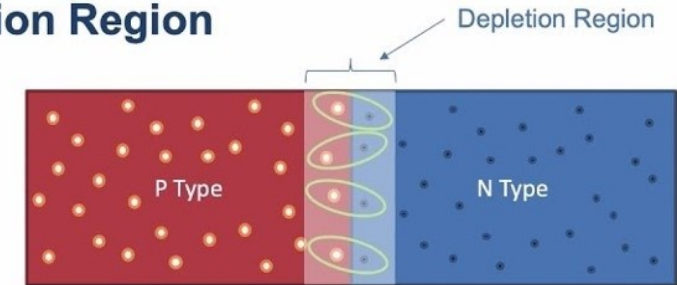


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P-N Junction: A Diode

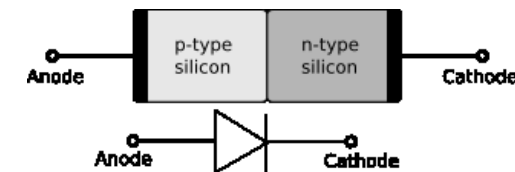
- When the p-doped semiconductor and the n-doped semiconductor come into contact, they start forming a region in their contact which has the opposite characteristics from their doping type.
- This region is formed because electrons move from the n-doped contact region to fill the (electron) holes in the p-doped contact region, and vice versa, which results in a lack of charge carriers in this region.
- This region is called depletion region, because the electrons (majority charge carriers in the n-doped semiconductor) are depleted in the part of this region on the n-doped semiconductor, and the holes (majority charge carriers in the p-doped semiconductor) are depleted in the part of this region on the p-doped semiconductor.

The Depletion Region



- Due to **diffusion**, electrons and holes will move around, and combine to form 'complete' atoms.
- This forms a region where there are **no free electrons**.
- This is called the **depletion region**.

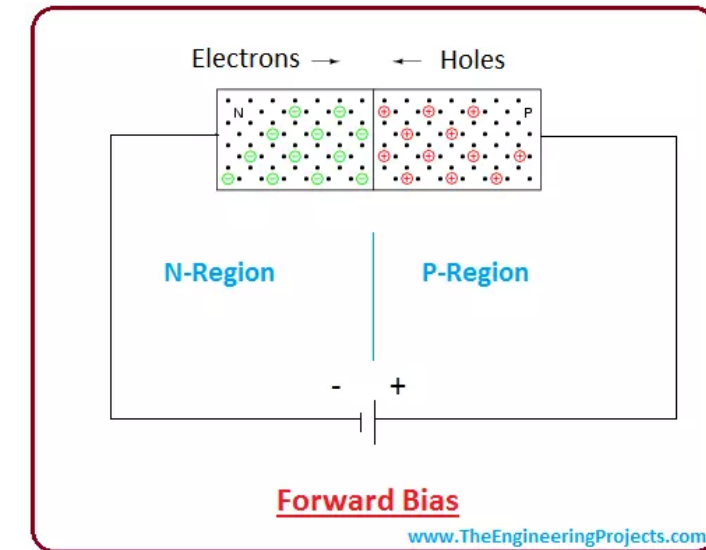
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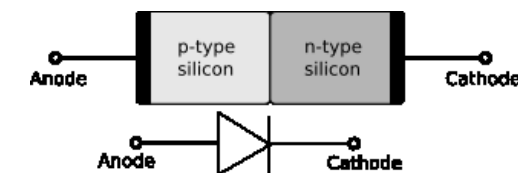
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P-N Junction: A Diode

- The depletion region along with the inherent properties of the p-doped and the n-doped semiconductor cause the p-n junction to act as a diode, which will allow current to pass when it is forward-biased and not allow current to pass when it is reverse-biased.
- Forward bias essentially provides a force for the majority carriers of each doped region to move towards the contact region, creating an electrical current, and narrowing the depletion region.



<https://images.theengineeringprojects.com/image/webp/2018/05/pn-junction.png.webp?ssl=1>



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